POLITECNICO MILANO 1863 | DEPARTMENT OF ELECTRONICS INFORMATION AND BIOENGINEERING

Borsa di Ricerca (2025)

DDD-DRC

01.10.2025 I Giuseppe Fiengo

Table of Contents (p1)

01

Intro & Motivation

- Problem Statement
- Context & Big Picture
- Motivation
- Research Objectives

02

Problem Formulation

- MBD problems
- Disturbances

03

State of the Art

- Comparative tables

04

Methodology

- Core idea
- Estimation Errors
- Failed Attempts
- Variants

Table of Contents (p2)

05

Experiments

- Evaluation metrics
- Preliminary results
- Comparison with baselines
- Error analysis / Failure cases

06

Discussion

- Interpretation of results
- Limitations of the method
- Open questions

07

Conclusion

- Key contributions
- Research roadmap

List of Acronyms

List of symbols

Timeline

Research & SOTA

I map relevant methods and results, highlight recurring limitations, and formulate working hypotheses. Output: comparative table and 3–5 testable questions.

Setup

I make the environment reproducible (data, seeds, metrics) and align the baselines. Output: end-to-end script that runs in 1 command and sanity plots.

Methodology

I define the algorithm/architecture with justified choices and an ablation plan. Output: schema, pseudocode, success criteria, and explicit assumptions.

Experiments

I run controlled experiments with metrics and CIs, comparing them with baselines, and analyzing errors. Output: tables, graphs, and labeled failure cases.

Conclusion

I summarize the contributions, limitations, and practical implications; I propose the next roadmap. Output: executive summary and decision checklist.

Points for next meeting

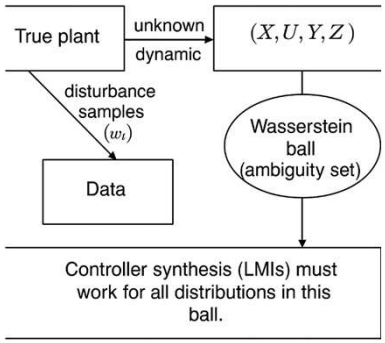
Ongoing:
- MoVe Server
- MATLABT

- Reviewed the code in the buswayne/in-context-bldc and buswayne/ST-microelectronics repositories
- Done some local training to test the speed controller files
- Extensive Research of the SOTA
- Got the department VPN

Intro & Motivation

01

Problem Statement



Problem:
Modern control systems must operate under **poorly known disturbance distributions**. Traditional robust and stochastic methods assume either deterministic uncertainty sets or fully known probability models. Both assumptions break in realistic high-data, low-trust industrial environments.

Need:
A controller that guarantees stability and performance **for all disturbance distributions compatible with available data**, without requiring parametric identification or Gaussian assumptions.

Gap:
State-of-the-art *model-based* distributionally robust control (DRC) still requires the **full plant matrices** A, B_u, B_w, C, D.
In industrial practice, these are rarely known and often estimated unreliably from short, noisy experiments.

Context & Big Picture

01.1

Industry

- Aerospace, energy, autonomous systems, industrial automation: all rely on feedback under uncertainty and all face poorly known disturbance distributions.
- Modern systems are instrumented heavily; data availability is rarely the bottleneck. The bottleneck is using this data rigorously.
- Distributionally robust control fits the "high-data, low-trust" industrial setting: we measure disturbances, but we don't trust any parametric distribution.

Problem class formalization

The paper solves a worst-case stochastic performance problem:

$$\sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \, \|z_t\|^2$$

subject to:

- LTI plant,
- output-feedback controller,
- disturbances in a 2-Wasserstein ball around a nominal Gaussian,
- correlated or independent disturbance models.

They show this can be rewritten as a **robust H_2** synthesis problem and solved through LMIs

Motivation (Why this research matters)

01.2

Why Direct Data-Driven method for Distributionally Robust Control applications?

Right now, the baseline paper assumes the full plant matrices $A, B_w, B_u, C_z, \ldots$are known. In practice, they rarely are.

Industries that care about robustness usually operate on black-box or partially identified systems. Making DRC *directly* data-driven removes the entire modelling pipeline:

- No system identification bias.
- No parametric model error contaminating the Wasserstein radius.
- No "model-based but data-estimated" contradiction.

Uncertainty is represented *exactly* where it enters: disturbance samples and empirical moment structure. Direct data removes one full layer of approximation. The ambiguity set becomes tied to sample distributions instead of a presumed nominal Gaussian, and the LMIs operate directly on empirical second-moment constraints.

The model-based version is elegant mathematically. A data-driven version would be operationally relevant.

Why not the usual methods?

- **Classical robust control** assumes deterministic uncertainty sets (polytopes, norm-bounded blocks). These give worst-case designs that are often overly conservative and completely ignore distributional structure.
- **Standard stochastic control** assumes perfect knowledge of the disturbance distribution. In practice, you get only samples and you never know the true distribution.
- **Scenario methods** give sample-based bounds but scale poorly and lack closed-form LMIs.
- **Gaussian assumptions** break immediately if the disturbance has heavy tails, asymmetry, or nonlinear correlations.

Distributionally robust control fills the gap: it handles *sets of distributions* instead of a single guessed one.

Control side

- The Wasserstein ball gives performance guarantees that hold for any disturbance distribution within a geometry-constrained class.
- LMIs give computational tractability.
- Output-feedback DRC closes a gap: previous work mainly handled state-feedback or finite-horizon versions.
- The duality with robust H_2 control reveals that DRC is not some exotic new control theory. It's classical robust control with a structured uncertainty block.

DRC is implicitly data-driven. A direct formulation removes the remaining model-dependence.

Research objectives

01.3

- **Remove the need for explicit model identification**
  Avoid estimating the plant and incorporate only empirical quantities directly in the LMIs.
- **Embed empirical disturbance structure directly into the ambiguity set**
  Replace parametric covariances and assumed Gaussians with real disturbance samples and empirical matrices.
- **Develop a convex, implementable controller synthesis method**
  Maintain the tractability of the baseline's robust LMIs while removing the model-based dependency.
- **Ensure performance and stability guarantees under distributional uncertainty**
  Leverage Wasserstein-ball robustness without relying on model-based nominal dynamics.
- **Design and test a hybrid data–model pipeline**
  Including:
  - selective-run dataset formation (medoid method),
  - LPV linearization,
  - one-step residual decomposition,
  - uncertainty budget allocation to evaluate real feasibility.

Problem Formulation

02

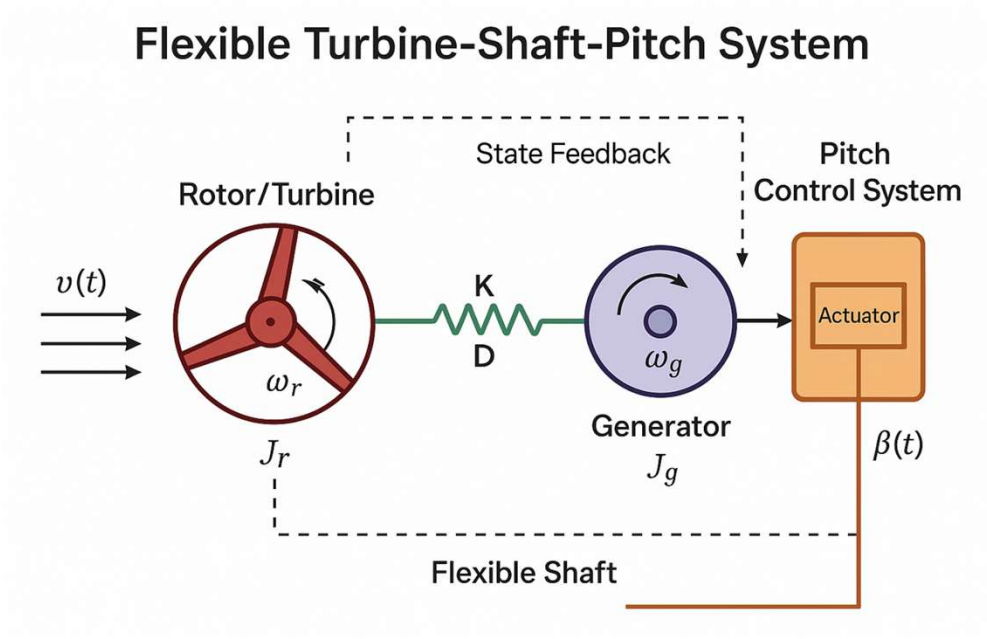
Reference Study Case System

$$\dot{x}(t) = A_{CT}x(t) + B_{CT}u(t) + E_{CT}w(t),$$

$$A_{CT} = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_{CT} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_{CT} = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Assumption 1:
Known System Model.



MBD problems

02.1

Objective & Formulation

This cost **does not** force the system's states to converge to zero.
Instead, it measures the **steady-state energy** of states and inputs under disturbances.

- In the absence of disturbances, minimizing this cost would yield a standard LQR-type behavior where the controller drives $x_t \to 0$.
- In the DRC setting, disturbances are always present and distributionally uncertain, so the states cannot be driven to zero.

The objective becomes:
minimize the average (RMS) state and control energy under the worst disturbance distribution inside the Wasserstein ambiguity set.

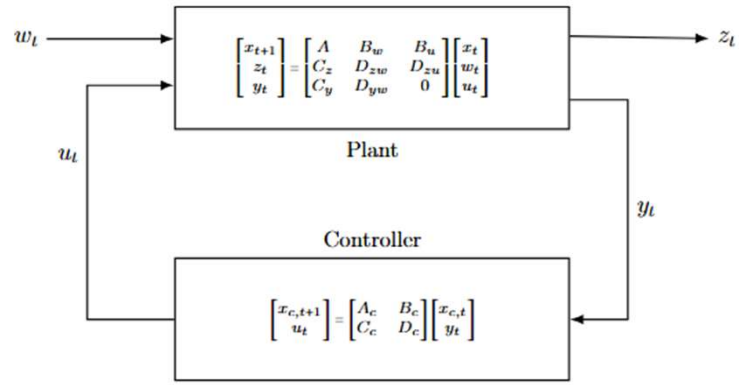


Figure 1.1: Closed-loop system with y feedback.

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathcal{A} \quad \text{isstable}$$

$$J(\theta) = \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|z_t\|^2$$

$$\text{s.t.} \quad \begin{aligned} \chi_{t+1} &= \mathcal{A}\chi_t + \mathcal{B} w_t, \\ z_t &= \mathcal{C}\chi_t + \mathcal{D} w_t \end{aligned}, \qquad \chi_0 = 0$$

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0$$

The performance objective penalizes how much the system amplifies disturbances—not how close the trajectory gets to the origin.

LMI-MBD problem

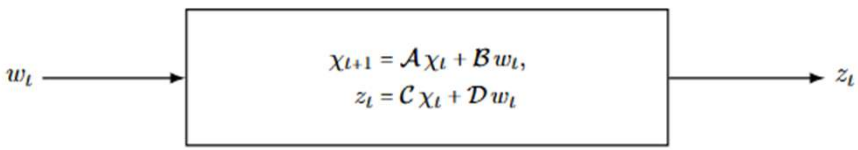


Figure 1.2: Lifted closed loop with state χ_t and matrices $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$.

Introducing decision variables

$$\lambda \geq 0, \quad Q, \quad X = X^\intercal, \; Y = Y^\intercal, \; K, L, M, N,$$

and defining

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

The objective bacame:

$$\min_{\lambda \geq 0,\; Q,\; X = X^\intercal,\; Y = Y^\intercal,\; K, L, M, N} \operatorname{trace}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2$$

subject to a block LMI that depends on the ambiguity model.

Correlated model ($\mathcal{W}_{\text{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\intercal & \mathbf{C}^\intercal \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\intercal & \mathbf{D}^\intercal \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} < 0, \quad \mathbb{P} \geq 0.$$

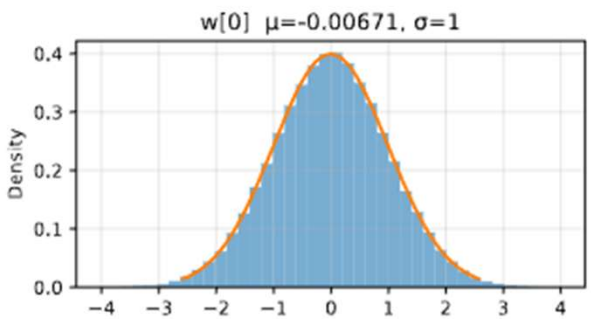
Independent model ($\mathcal{W}_{\text{ind}}$). Impose the pair of LMIs

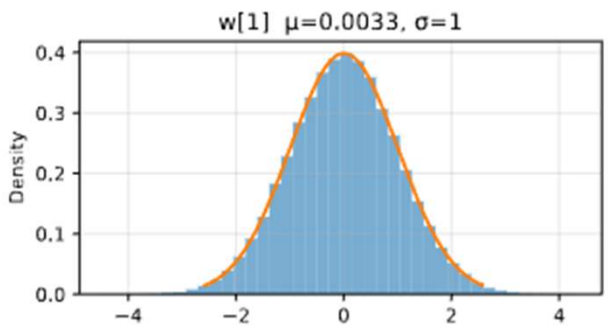
$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\intercal & \mathbf{C}^\intercal \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} < 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\intercal & \mathbf{D}^\intercal \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} < 0, \quad \mathbb{P} \geq 0.$$

Disturbance

02.2

Assumption 2:
Zero-mean nominal distribution with Identity covariance.





Algorithm 1 Sampling Gaussian disturbances inside a 2-Wasserstein ball

Require: nominal covariance $\Sigma_{\text{nom}} \in \mathbb{S}_{++}^{n_w}$, radius $\gamma > 0$, time horizon T, mode $\texttt{mode} \in \{\text{independent}, \text{correlated}\}$, correlation coefficient $\rho \in (0,1)$

Choose a candidate covariance $\Sigma_{\text{raw}} \in \mathbb{S}_{++}^{n_w}$, e.g. $\Sigma_{\text{raw}} = \begin{bmatrix} 0.69 & 0.53 \\ 0.53 & 1.61 \end{bmatrix}$
Compute $d_{\text{raw}} \leftarrow W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))$
if $d_{\text{raw}} \le \gamma$ **then**
  $\Sigma_{\text{use}} \leftarrow \Sigma_{\text{raw}}$
else
  Compute $S \leftarrow \Sigma_{\text{raw}}^{1/2}$, $A \leftarrow (S\Sigma_{\text{nom}}S)^{1/2}$
  Define $\Sigma(t) \leftarrow ((1-t)S + tA)^2$ for $t \in [0,1]$
  Find $t^\star \in (0,1]$ by bisection such that $W_2(\mathcal{N}(0, \Sigma(t^\star)), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma$
  $\Sigma_{\text{use}} \leftarrow \Sigma(t^\star)$
if $\texttt{mode} = \text{independent}$ **then**
  Compute Cholesky factor $\Sigma_{\text{use}} = LL^\top$
  for $t = 0$ to $T-1$ **do**
    Draw $z_t \sim \mathcal{N}(0, I_{n_w})$
    $w_t \leftarrow Lz_t$
else ▷ correlated AR(1) case
  $F \leftarrow \rho I_{n_w}$
  $Q \leftarrow \Sigma_{\text{use}} - F\Sigma_{\text{use}}F^\top$
  Compute Cholesky factors $\Sigma_{\text{use}} = L_\Sigma L_\Sigma^\top$, $Q = L_Q L_Q^\top$
  Draw $z_0 \sim \mathcal{N}(0, I_{n_w})$ and set $w_0 \leftarrow L_\Sigma z_0$
  for $t = 0$ to $T-2$ **do**
    Draw $\eta_t \sim \mathcal{N}(0, I_{n_w})$
    $\varepsilon_t \leftarrow L_Q \eta_t$
    $w_{t+1} \leftarrow Fw_t + \varepsilon_t$
return $\{w_t\}_{t=0}^{T-1}$

Algorithm 2 Plotting marginal ellipses and Wasserstein bounds

Require: samples $w \in \mathbb{R}^{T \times n_w}$, nominal covariance Σ_{nom}, Wasserstein radius γ, coordinate pair (i, j), confidence level k (e.g. $k = 2$)

1: Extract 2D samples $W = w[:, (i, j)]$
2: Compute empirical mean μ and sample covariance Σ_{emp}
3: Compute 2D nominal covariance $\Sigma_{\text{nom},2d} = \Sigma_{\text{nom}}[(i, j), (i, j)]$
4: Generate a small set of boundary covariances:

$$\{\Sigma_b\} \leftarrow \text{axis-wise eigenvalue shifts of } \Sigma_{\text{nom}} \text{ by } \pm\gamma$$

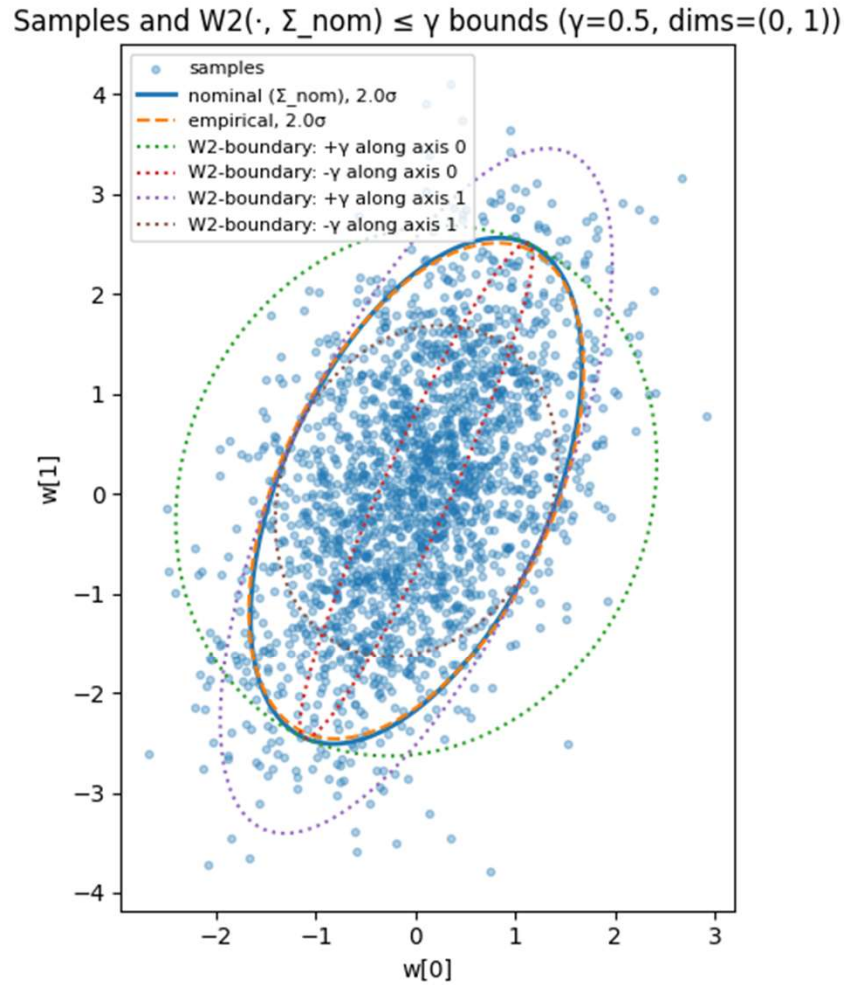
5: For each covariance $\Sigma \in \{\Sigma_{\text{nom},2d}, \Sigma_{\text{emp},2d}, \Sigma_b^{(i,j)}\}$:

1. Compute eigen-decomposition $\Sigma = V \Lambda V^\top$
2. Generate ellipse points

$$\mathcal{E}_k = \left\{ \mu + V \Lambda^{1/2} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} k : \theta \in [0, 2\pi] \right\}$$

3. Add to the plot using:
   - solid line for nominal Σ_{nom}
   - dashed line for empirical Σ_{emp}
   - dotted lines for boundary ellipses Σ_b

6: Scatter-plot the sample points W
7: Return rendered figure



The Wasserstein constraint is a **constraint on the covariance structure**, not on individual samples. Thus, **not** every sample lies inside these ellipses.

State of the Art

03

Where We Stand and What's Missing

Current data-driven and distributionally robust control methods provide powerful tools for dealing with uncertainty, noisy trajectories, and model ambiguity.

- Behavioral approaches based on Willems' Fundamental Lemma describe the complete set of trajectories compatible with past data.
- S-lemma techniques characterize all models consistent with noise bounds and deliver non-conservative robust LMI conditions.
- Extended-state methods show how dynamic controllers can, in principle, be embedded directly into data.
- Distributionally robust optimization gives principled ambiguity sets—Wasserstein, ℓ_1 ,cost-aware—together with finite-sample guarantees.

Yet **none of these frameworks fully addresses the problem studied here**. Almost all require either **state-feedback**, **known controller dynamics**, or **disturbance models incompatible with Wasserstein ambiguity**. Some assume noise is measurable or enters the system as part of the input; others work on probability simplices rather than continuous disturbance channels. None provides a convex, data-driven formulation for **dynamic output-feedback** where the controller has its own **unknown internal state**, and where the disturbance enters through explicit (B_w, D_{yw}, D_{zw})channels.
In short, the existing methods each solve *part* of the puzzle—but **no method simultaneously supports**:

- data-driven estimation of system maps;
- robust synthesis of **dynamic** output-feedback controllers;
- explicit disturbance channels;
- **Wasserstein-based distributional robustness**;
- and a **fully convex LMI formulation**.

This gap defines the motivation for the proposed framework.

Comparative table (methods vs limitations)

03.1

De Persis & Tesi (Formulas for data-driven control)

Pros for my objective

Gives explicit **data-driven formulas** for closed-loop matrices that plug into **Lyapunov LMIs**.

Shows how to combine **Willems' lemma + LMIs** in a clean way.

Provides **stabilization, LQR and robustness** in a single unified data-driven framework.

Conceptually close to what I want (direct data-driven + LMIs).

Limitations / why not directly usable

Core results are mainly for **state-feedback**; dynamic output-feedback is more restricted and structurally different from my controller.

Noise/uncertainty is modeled via **quadratic constraints on data**, not via **Wasserstein ambiguity** on the disturbance distribution.

Disturbance channels are not separated as $(B_u, B_w, D_{yw}, D_{zw})$, so it does not match the disturbance–explicit robust LMI structure.

Would require a substantial reformulation to handle: dynamic output-feedback, internal controller state, and Wasserstein disturbance uncertainty simultaneously.

LMI = Linear Matrices Inequality; LQR = Linear Quadratic Regulator

van Waarde et al. (From noisy data to feedback controllers via a matrix S-lemma)

Pros for my objective

Very **sharp handling of noise**: characterizes all (A, B) consistent with noisy data via a **QMI**.

Uses a **matrix S-lemma** to get **nonconservative robust LMIs** (no nasty conservatism from over-bounding).

Fixed-size LMIs (dimension does not grow with data length).

Very good conceptual template for "data → model set → robust LMI".

Limitations / why not directly usable

Only considers **static state-feedback** $(u = Kx)$;no dynamic output-feedback with internal controller state.

Noise is modeled as **bounded additive process noise**, not as a random variable in a **Wasserstein ambiguity set**.

No explicit disturbance channel (B_w) and no performance outputs with (D_{yw}, D_{zw}) like in my setup.

Would need an entirely new lifting to handle my **distributionally robust** disturbance model and dynamic controller parametrization.

QMI = Quadratic Matrice Inequality

Ravikkumar (An extended state framework for direct data-driven control)

Pros for my objective

Shows how to build an **extended state** that includes the controller and use WFL* on that extended system.

Demonstrates that dynamic controllers can, in principle, be handled **purely from data** when the extended state is observable from trajectories.

Conceptually close to my "plant + controller as one system" viewpoint.

Useful as a mental model for embedding controller dynamics in data-driven design.

Limitations / why not directly usable

Assumes the **controller structure is known** so the extended state can be reconstructed; in my case, the controller matrices are unknown decision variables.

Essentially relies on **noise-free (or very mildly noisy)** data; noise is not modeled via Wasserstein or general distributional ambiguity.

I cannot generate open-loop trajectories of the internal controller state (x_c), so I cannot apply the same extended-state behavioral representation.

Not formulated as a **robust LMI** with explicit disturbance channels and distributionally robust constraints.

WFL = Willem's Fundamental Lemma

Schuurmans & Patrinos (Data-driven DRO of partially observable MJLS)

Pros for my objective

Provides a **full pipeline**: data → ambiguity sets → **distributionally robust controller**, in an output-feedback setting.

Deals with **partial observability** and estimation + control together.

Shows how to embed probability ambiguity into robust control design.

Illustrates how data-driven ambiguity can shrink with more data (adaptive robustness).

Limitations / why not directly usable

Uncertainty is on **mode transition probabilities of a Markov jump system**, not on an additive disturbance entering via (B_w, D_{yw}, D_{zw}).

Ambiguity sets are **(ℓ_1)-balls on probability simplices**, not **Wasserstein balls on continuous-valued disturbances**.

Model structure is MJLS with mode-dependent dynamics; this does not match my single LTI plant with robust disturbance channels.

To reuse it, I would essentially need to reformulate my system as MJLS or abandon the Wasserstein disturbance model, which breaks my main objective.

MJLS = Markov Jump Linear System; LTI = Linear Time Invariant

Esfahani & Kuhn (Wasserstein DRO foundations)

Pros for my objective

Gives the **theoretical backbone** for using **Wasserstein balls** as ambiguity sets around empirical distributions.

Provides **finite-sample performance guarantees** and asymptotic consistency for Wasserstein DRO.

Shows how Wasserstein DRO induces **regularization terms** in the optimization.

Justifies the **scaling of the Wasserstein radius** (γ) w.r.t. sample size and dimension.

Limitations / why not directly usable

Framework is for **static optimization problems** $\min_x \sup_{Q \in U} E_Q\,[h(x,\xi)]$, not dynamic output-feedback controllers with LMIs.

No explicit state-space model with **disturbance channels**; disturbances appear as generic random variables (ξ) in the cost.

To use it directly, I would need to rewrite my whole LMI stability/performance conditions as one static DRO objective, which is not natural for my problem.

Does not give a ready-made **LMI** or controller structure; it's a theoretical tool that I have to adapt manually into my robust LMI framework.

DRO = Distributionally Robust Optimisation

Schuurmans & Patrinos (Cost-aware DRO / CADRO)

Pros for my objective

Shows that ambiguity sets can be **shaped using the cost function**, not only symmetric balls (e.g. Wasserstein, (ℓ_p)).

Reduces **conservatism** compared to vanilla metric balls by biasing ambiguity along "cost-sensitive" directions.

Conceptually suggests that a **cost-aware Wasserstein radius** or anisotropic ambiguity could be beneficial in my LMI design.

Useful inspiration for designing less conservative ambiguity sets in control.

Limitations / why not directly usable

Ambiguity sets are defined on **finite-dimensional probability vectors**, not on continuous disturbance distributions driving a state-space system.

Again, a **static DRO** setup (decision variable + probability vector), not a dynamic LTI system with output-feedback and Lyapunov LMIs.

No direct recipe to produce **matrix uncertainty sets** for (A, B_u, B_w) in LMI form; it operates at the probability level, not the system level.

To use it, I would need to derive an entirely new mapping from "cost-aware ambiguity" to **closed-loop LMI constraints**; nothing in the paper does that.

Methodology

04

DDD using rho as decision variables -> Can't make this convex
DDD via Willem's Lemma -> Can't define the LMI matrices
DDD using Estimation Errors -> Convex but too strict solution's space

Core idea (pseudo-algorithms)

04.1

Assumption 1:
Known nominal covariance of the noise and the worst-case radius of the Wasserstein-ball to be $\gamma = 0.5$.

if unknown disturbance covariance **then**

Choose smallest n_w with

$$\sum_{i=1}^{n_w} s_i \geq \eta \sum_{i=1}^{n} s_i$$

Construct

$$\hat{B}_w \, \widehat{\Sigma}_w^{1/2} = U_s(:, 1{:}n_w) \, \mathrm{diag}(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}})$$

Compute

$$\Sigma_{\mathrm{nom}} = \tfrac{1}{2}\Big(\tfrac{1}{T}\hat{W}\hat{W}^\top + (\tfrac{1}{T}\hat{W}\hat{W}^\top)^\top\Big) + \varepsilon I$$

Algorithm 2 One–Step Linear Regression (SysID) and Residual Decomposition

Input: Representative dataset (X_-, U_-, X_+)

Parameters: regularization $\rho \geq 0$, energy fraction η, known/unknown covariance mode

(A) One-step regression

Form regression matrix $D_x = [X_-; U_-]$

Compute regularized right inverse:

$$D_x^\dagger = D_x^\top (D_x D_x^\top + \rho I)^{-1}$$

Estimate nominal dynamics:

$$[\,\hat{A} \; \hat{B}_u\,] = X_+ D_x^\dagger$$

(B) Residual computation

Compute state residuals:

$$R = X_+ - \hat{A} X_- - \hat{B}_u U_-$$

Residual covariance:

$$S_R = \tfrac{1}{T} R R^\top, \qquad \tilde{S}_R = \tfrac{1}{2}(S_R + S_R^\top) + \varepsilon I$$

Eigen-decomposition:

$$\tilde{S}_R = U_s \, \mathrm{diag}(s_1, \ldots, s_n) \, U_s^\top$$

with $s_1 \geq \cdots \geq s_n$.

(C) Disturbance-channel extraction

Use known $\Sigma_{\mathrm{nom}} = Q \mathrm{diag}(\lambda_i) Q^\top$

$n_w = \dim(\Sigma_{\mathrm{nom}})$

$U_p = U_s(:, 1{:}n_w)$

Build matched disturbance channel:

$$\hat{B}_w = U_p \, \mathrm{diag}(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}) \, \Sigma_{\mathrm{nom}}^{-1/2}$$

Output:

Nominal $(\hat{A}, \hat{B}_u)$, disturbance channel $\hat{B}_w$, residuals R, nominal covariance Σ_{nom}

Medoid Selection

Algorithm 1 Dataset Preprocessing and Medoid Selection

Input: N trajectories $\Gamma_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}$, time lengths T_i
Parameters: PE threshold $\varepsilon_{\mathrm{PE}}$, scaling offset ε, weights w_k
(A) Truncate and PE filtering
$T \leftarrow \min_i T_i$
for $i = 1$ to N **do**
  Truncate each Γ_i to horizon T
  Sample $x_0^{(i)} \sim \mathcal{N}(0, I)$
  Form $\Phi_i = [X_i; U_i]$
  if $\sigma_{\min}(\frac{1}{T}\Phi_i\Phi_i^\top) \leq \varepsilon_{\mathrm{PE}}$ **then**
    discard Γ_i
(B) Normalization across datasets
for each block $k \in \{X, U, Y, Z, X_+\}$ **do**
  Compute scale $s_k = \frac{1}{N}\sum_{i=1}^{N} \|\Gamma_i[k]\|_F + \varepsilon$
  for $i = 1$ to N **do**
    $\tilde{\Gamma}_i[k] \leftarrow \Gamma_i[k]/s_k$
(C) Medoid selection
for $i = 1$ to N **do**
  Compute total normalized Frobenius distance:

$$d_i = \sum_k w_k \sum_{j=1}^{N} \|\tilde{\Gamma}_j[k] - \tilde{\Gamma}_i[k]\|_F^2$$

$i^\star = \arg\min_i d_i$
(D) Output
Representative dataset:

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+})$$

Algorithm on *Estimation Errors*

04.2

Step 1: Bounds on estimation errors

$$\begin{bmatrix} A & B_u \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} + \Delta_x, \quad \Delta_x = \begin{bmatrix} \Delta_A & \Delta_B \end{bmatrix},$$

$$\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_A + E_B,$$

Bound on the perturbation block E. We start from the base bound

$$\beta = RD^{\dagger} \in \mathbb{R}^{(n_x+n_u)\times(n_x+n_u)},$$

and split it between the state and input parts as

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x+n_u}}, \qquad \beta_B = \beta\sqrt{\frac{n_u}{n_x+n_u}}.$$

The corresponding spectral–norm bounds are

$$\kappa_A := \|\beta_A\|_2, \qquad \kappa_B := \|\beta_B\|_2,$$

and we assume

$$\|\Delta_A\|_2 \le \kappa_A, \qquad \|\Delta_B\|_2 \le \kappa_B.$$

The lifted perturbation block is

$$E = \begin{bmatrix} \Delta_A Y + \Delta_B M & \Delta_A + \Delta_B N C_y \\ 0 & X\Delta_A \end{bmatrix},$$

where the design matrices satisfy the norm bounds

$$\|Y\|_2 \le t_Y, \qquad \|M\|_2 \le t_M, \qquad \|N\|_2 \le t_N, \qquad \|X\|_2 \le t_X,$$

and C_y is constant with $c_y := \|C_y\|_2$.

Using submultiplicativity and the triangle inequality, we obtain

$$\|\Delta_A Y + \Delta_B M\|_2 \le \|\Delta_A\|_2\|Y\|_2 + \|\Delta_B\|_2\|M\|_2 \le \kappa_A t_Y + \kappa_B t_M =: e_{11},$$
$$\|\Delta_A + \Delta_B N C_y\|_2 \le \|\Delta_A\|_2 + \|\Delta_B\|_2\|N\|_2\|C_y\|_2 \le \kappa_A + \kappa_B t_N c_y =: e_{12},$$
$$\|X\Delta_A\|_2 \le \|X\|_2\|\Delta_A\|_2 \le t_X\kappa_A =: e_{22}.$$

A conservative global bound on E follows from

$$\|E\|_2^2 \le \|E\|_F^2 \le e_{11}^2 + e_{12}^2 + e_{22}^2 =: \gamma_E,$$

which implies

$$E^{\top}E \preceq \beta_E \quad \text{with} \quad \beta_E := \gamma_E I = \left(e_{11}^2 + e_{12}^2 + e_{22}^2\right) I.$$

Step 2

Robustification of the Lyapunov inequality under model error.
Starting from

$$M(\mathbf{A}_{\text{true}}, P) = \begin{bmatrix} -P & \mathbf{A}_{\text{true}}^{\top} \\ \mathbf{A}_{\text{true}} & -P \end{bmatrix} \prec 0 \quad \Longleftrightarrow \quad -P + \mathbf{A}_{\text{true}}^{\top} P^{-1} \mathbf{A}_{\text{true}} \prec 0,$$

insert the decomposition

$$\mathbf{A}_{\text{true}} = \mathbf{A} + E, \qquad E = E_A + E_B.$$

Expanding gives

$$-P + \mathbf{A}^{\top} P^{-1} \mathbf{A} + E^{\top} P^{-1} E + E^{\top} P^{-1} \mathbf{A} + \mathbf{A}^{\top} P^{-1} E \prec 0.$$

Bounding the inverse metric. Introduce $\sigma_p > 0$ such that

$$P^{-1} \prec \sigma_p I.$$

This is equivalent to

$$\begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0.$$

Thus

$$E^{\top} P^{-1} E \preceq \sigma_p E^{\top} E, \qquad E^{\top} E \preceq B_E \quad \Rightarrow \quad E^{\top} P^{-1} E \preceq \sigma_p B_E.$$

Bounding the mixed term. For any $S \succ 0$,

$$E^{\top} P^{-1} \mathbf{A} + \mathbf{A}^{\top} P^{-1} E = 2\langle E,\ P^{-1}\mathbf{A} \rangle \ \preceq\ E^{\top} S E + \mathbf{A}^{\top} P^{-1} S^{-1} P^{-1} \mathbf{A}.$$

Introduce $\sigma_s > 0$ with

$$S \prec \sigma_s I \quad \Longleftrightarrow \quad \sigma_s I - S \succ 0.$$

Then

$$E^{\top} S E \preceq \sigma_s B_E.$$

Thus:

$$-P + \mathbf{A}^{\top}\left(P^{-1} - P^{-1} S P^{-1}\right)\mathbf{A} + \sigma_p B_E + \sigma_s B_E \prec 0.$$

Define

$$\bar{P} := P - \sigma_e B_E, \quad \sigma_e = \sigma_s + \sigma_p,$$

so $P_s \succ 0$. Its inverse satisfies

$$P_s^{-1} = P^{-1} + P^{-1} S P^{-1} \succ 0$$

Therefore

$$-\bar{P} + \mathbf{A}^{\top} P_s^{-1} \mathbf{A} \prec 0 \quad \Longleftrightarrow \quad \begin{bmatrix} -\bar{P} & \mathbf{A}^{\top} \\ \mathbf{A} & -P_s \end{bmatrix} \prec 0.$$

Auxiliary lifting. Define

$$G = \sigma_p I = G^\top \succ 0, \qquad G \succ P^{-1},$$

and introduce slack $H \succ 0$ via

$$\begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0 \quad \Longleftrightarrow \quad H - GS^{-1}G \succ 0.$$

Final condition. The robust requirement becomes

$$G + H - P_s^{-1} \succ 0,$$

equivalently,

$$\begin{bmatrix} G - H & I \\ I & P_s \end{bmatrix} \succ 0.$$

And thus, $P_s^{-1} = P^{-1} + P^{-1}S^{-1}P^{-1} \prec G + GS^{-1}G \prec G + H$.

Final program. The Young-relaxed data-driven synthesis (correlated case) is

$$\min_{\substack{\lambda \geq 0,\ Q \succeq 0, \\ S \succeq 0\ \ P_s \succeq 0\, H \succeq 0\ \ G = \sigma_p I = G^T \succeq 0 \\ \sigma_s \geq 0,\ \sigma_p \geq 0,\ \{t\}_{i \in (K,L,M,N,X,Y)} \\ X = X^T,\ Y = Y^T,\ K,\ L,\ M,\ N}} \quad \mathrm{tr}(Q\,\Sigma_{\mathrm{nom}}) + \lambda\gamma^2 + \sum_{i \in (K,L,M,N,X,Y)} \rho\,(t_i + \varepsilon)$$

$$\text{s.t. } P = P^T = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) \prec 0$$

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{1,1} \leftarrow -P + (\sigma_s + \sigma_p)\,\beta_E$$

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{n,n} \leftarrow -P_s$$

$$\begin{bmatrix} G & I \\ I & P \end{bmatrix} \succ 0, \qquad \begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0, \qquad \begin{bmatrix} G + H & I \\ I & P_s \end{bmatrix} \succ 0$$

$$\sigma_s I - S \succ 0\ , \quad \begin{bmatrix} t_j I_{n^j_{row}} & j \\ j^\top & t_j I_{n^j_{col}} \end{bmatrix}\Bigg|_{j \in (K,L,M,N,X,Y)} \succeq 0.$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are built from the nominal data-based maps $(\hat{A}, \hat{B}, \hat{C}_y, \hat{D}_{yw}, \hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw})$. For the correlated case, we have that $n = 4$, whereas in the independent one $n = 2$.

Failed Attempts

04.3

Willem's Fundamental Lemma

- A single persistently exciting trajectory uniquely spans all feasible finite trajectories of a controllable LTI system.
- If input u is **PE** of order $n + L$, then every length-L (u, y) trajectory lies in the column space of the Hankel data matrix:

$$\mathcal{B}_L(A, B, C, D) = \operatorname{im} \begin{bmatrix} H_L\left(u_{[0,T-1]}\right) \\ H_L\left(y_{[0,T-1]}\right) \end{bmatrix}.$$

Why use it?

- Avoids identifying the plant
- Provides linear parametrization of feasible trajectories
- Lets us build controller directly from data

What I tried:

- Embed WFL inside the robust output-feedback LMI synthesis
- Replace model matrices by Hankel parametrisation

Why it fails:

a) Noise breaks exact WFL membership:

We get: $\begin{bmatrix} u_p \\ y_p \end{bmatrix} g = \begin{bmatrix} u_p \\ y_p \end{bmatrix} + \delta,$

Noisy data -> no exact behavioral consistency

a) Tried to replace Hankel parametrization with FIR Toeplitz constraints:

But FIR imposes a controller structure, not the true plant behavior; in other words, "replacing the Hankel parametrization by an FIR controller removes the behavioral foundation entirely, it yields a different optimization problem that does not preserve the guarantees of the Willems Fundamental Lemma."

Regression-Based Identification

Identifying the plant using regularized LS

$$A := X_+ X^\top (XX^\top + \rho_x I)^{-1},$$
$$B_u := X_+ U^\top (UU^\top + \rho_u I)^{-1},$$
$$B_w := X_+ W^\top (WW^\top + \rho_w I)^{-1},$$

seems reasonable for:

- Regularized against noise
- Not fixed plant matrices
- Easy to implement

Good for Prediction, Not so much for LMI synthesis.

Why it fails:

a) The identified matrices contain inverses of Gram matrices
b) These inverse form bilinear terms inside the LMI constraints
c) No trick fixes it

I tried auxiliary variables, surrogate optimization, Shur complements, ...

Replacing the inverse will keep the bilinearities.

"Regression-based identification injects non-affine inverse operators inside the LMI constraints. This destroys convexity in an unavoidable way. Therefore, regression-driven identification cannot be embedded inside a convex distributionally robust LMI pipeline."

Variants / possible extensions (optional)

04.4

Non-convex approach

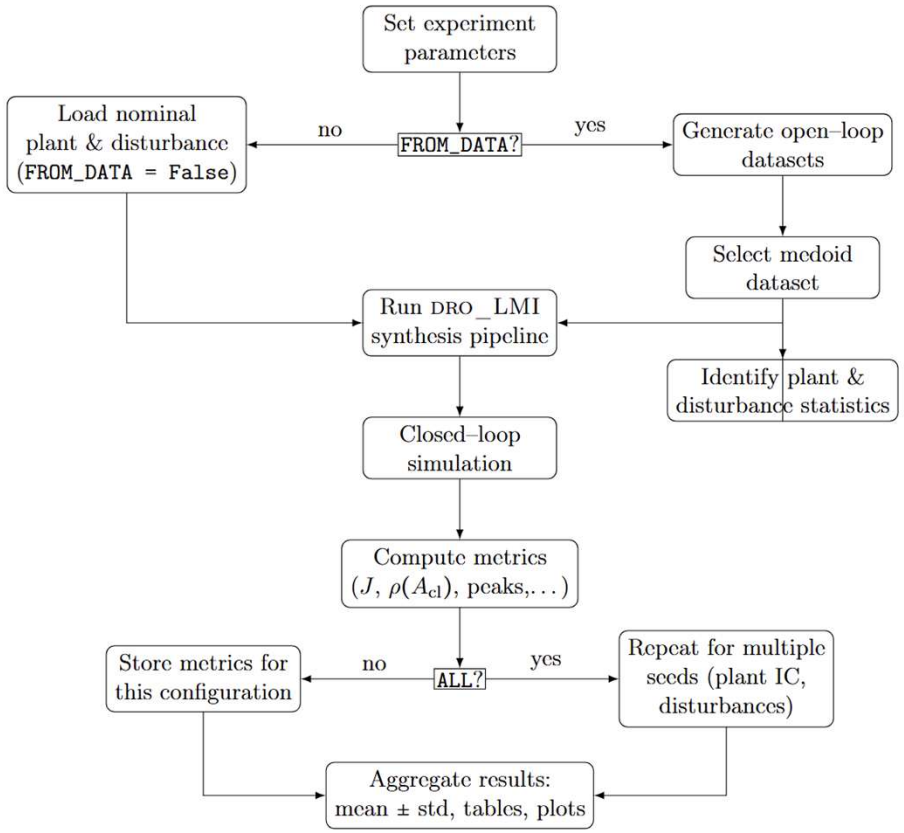
The Estimation Error approach fails due to too strict constraints and reduction of the Solution Space.

Thus, I could try to solve it initially non-convexly and with zero noise.

The latter makes the solutions to be unstable; the former is yet to be finalized in its implementation.

Experiments

05


Set experiment parameters
FROM_DATA?
no
yes
Load nominal plant & disturbance (FROM_DATA = False)
Generate open–loop datasets
Select medoid dataset
Run DRO_LMI synthesis pipeline
Identify plant & disturbance statistics
Closed–loop simulation
Compute metrics (J, $\rho(A_{\mathrm{cl}})$, peaks,...)
ALL?
no
yes
Store metrics for this configuration
Repeat for multiple seeds (plant IC, disturbances)
Aggregate results: mean ± std, tables, plots

Evaluation metrics

05.1

Simulation Setup

Given a state matrix $M \in \mathbb{R}^{n \times n}$:

1. Compute eigenvalues and eigenvectors:

$$MV = V\Lambda.$$

2. Select the eigenvalue with largest real part:

$$\lambda_* = \arg\max_{\lambda_i \in \sigma(M)} \Re(\lambda_i).$$

3. Extract the corresponding eigenvector v_*.
4. Build a real mode shape:

$$\tilde{v} = \frac{\Re(v_*)}{\|\Re(v_*)\|_2}.$$

5. Scale it to obtain the initial condition:

$$x_0 = r\,\tilde{v}, \qquad r \text{ small (e.g., } r = 0.05).$$

Why this is better than a random initial condition?

- Excites the **least damped / most critical mode** of the system.
- Produces a **structured and reproducible** test case.
- Avoids the noise and irrelevance of arbitrary random vectors.
- Gives a **meaningful transient** for evaluating controller performance.

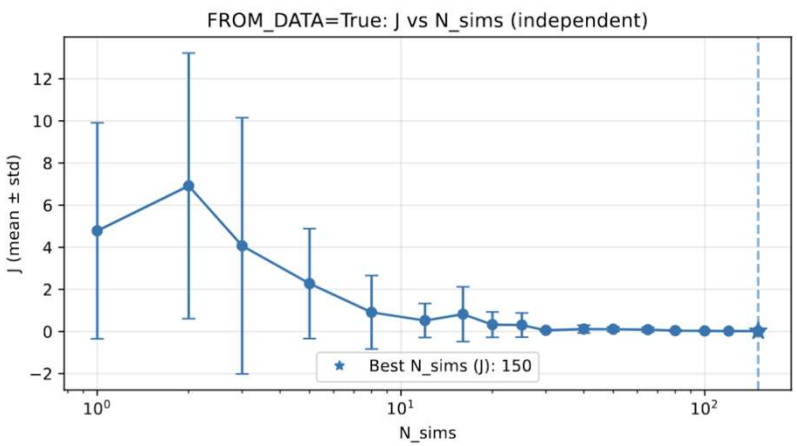
Category	Metric	Description
Performance	J	Finite–horizon quadratic cost on (z_k, x_k, u_k) over the simulation horizon.
Stability	$\rho(A_{\mathrm{cl}})$	Spectral radius of the closed–loop state matrix.
Performance	$z_{\max}$	Peak performance output, $z_{\max} = \max_k \|z_k\|_2$.
Performance	$u_{\max}$	Peak control effort, $u_{\max} = \max_k \|u_k\|_2$.
Steady–state	σ_z^{tail}	Standard deviation of z_k over the last portion of the horizon.
Steady–state	σ_y^{tail}	Standard deviation of y_k over the last portion of the horizon.
Computational	obj	Optimal objective value of the DRO_LMI synthesis problem.
Computational	t_{solve}	Wall–clock time required by the conic solver to compute the controller.
Computational	attempts	Number of solver attempts / restarts until a feasible solution is obtained.
Diagnostic	stress	Scalar diagnostic summarizing how "tight" or conservative the design is.
Aggregation	$\mu \pm \sigma$	Mean and standard deviation of each metric over all runs in `ALL` mode.

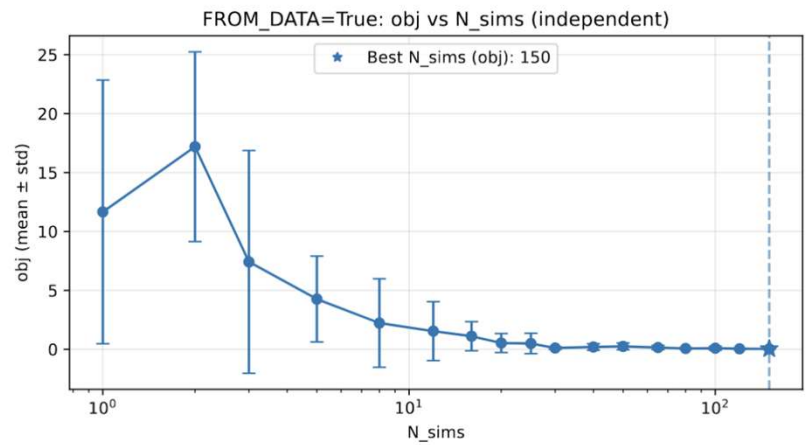
Table 2.1: Closed–loop performance and computational metrics used for comparison.

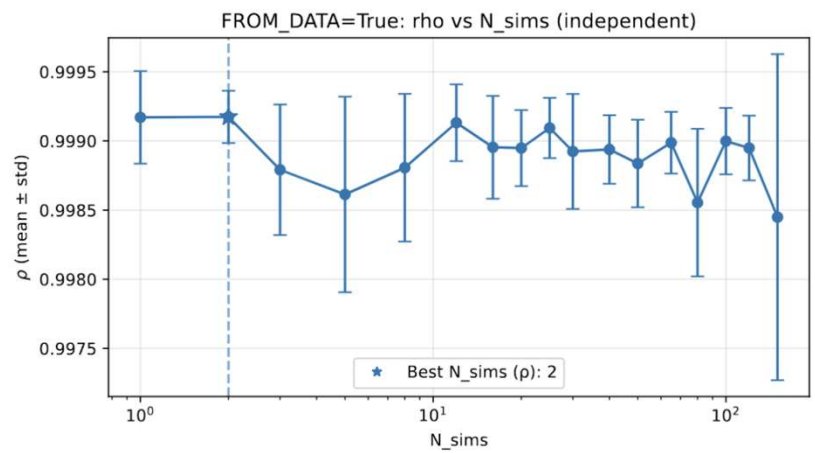
Assumption results

05.2

Medoid usage





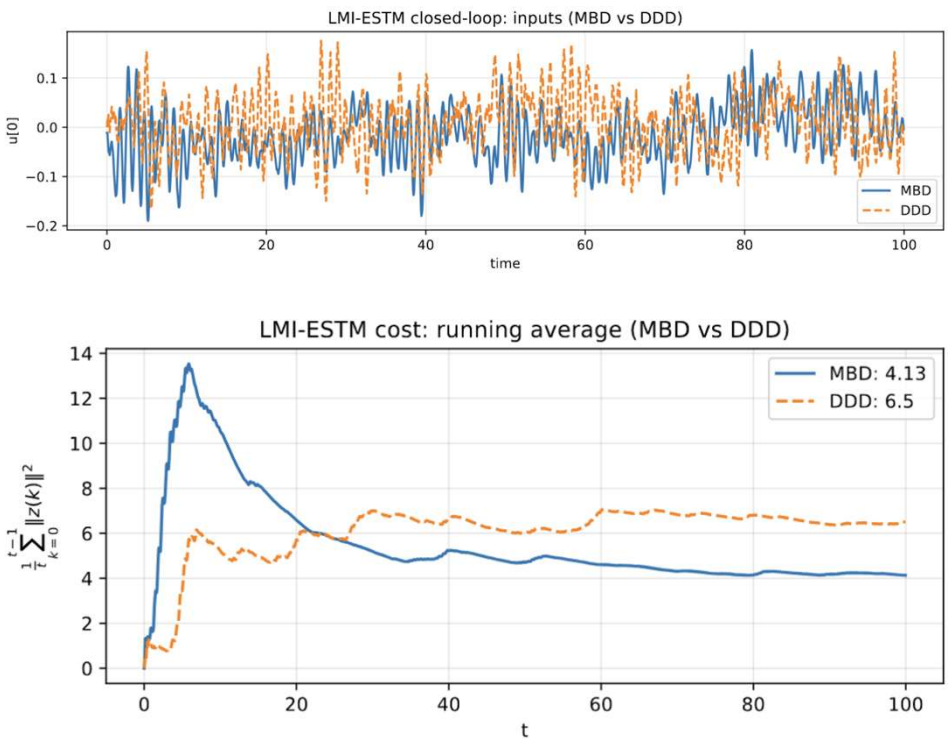


Comparison with baselines

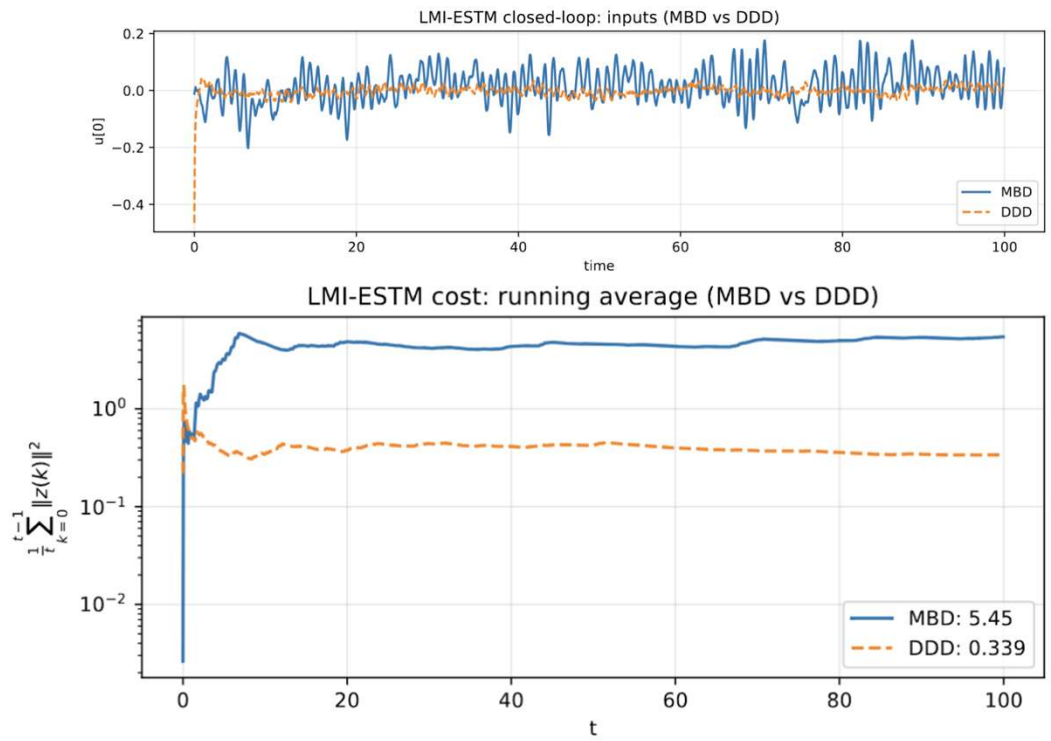
05.3

DD method

#N sim = 1



#N sim = 16

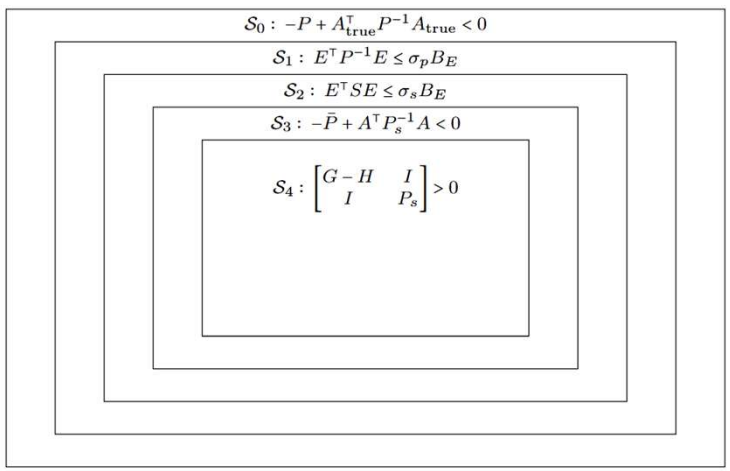


Comparison

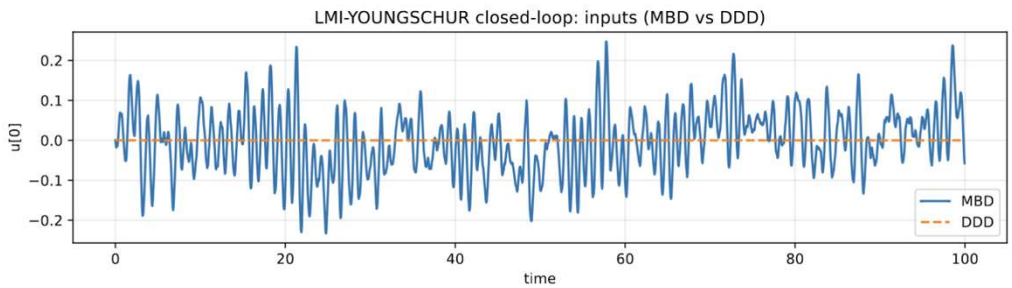
Metric	N sims = 1	N sims = 16	Comment
Objective value	16.504	0.927	Large drop; uncertainty shrinks with more data
λ_opt	21.999	1.234	Penalty on ambiguity reduces ~18×
trace(Q Σ)	10.999	0.618	Residual energy much smaller
γ (Wasserstein radius)	0.500	0.500	Same (input setting)
Spectral radius	0.999454	0.998622	Slight improvement in stability margin
Z cost	6.5035	0.3387	~19× improvement in closed-loop performance
Solver time [s]	0.36	1.58	Slightly slower with more data
Attempts	1	1	No change
Stress	2.7056	1.3051	Constraints far less strained

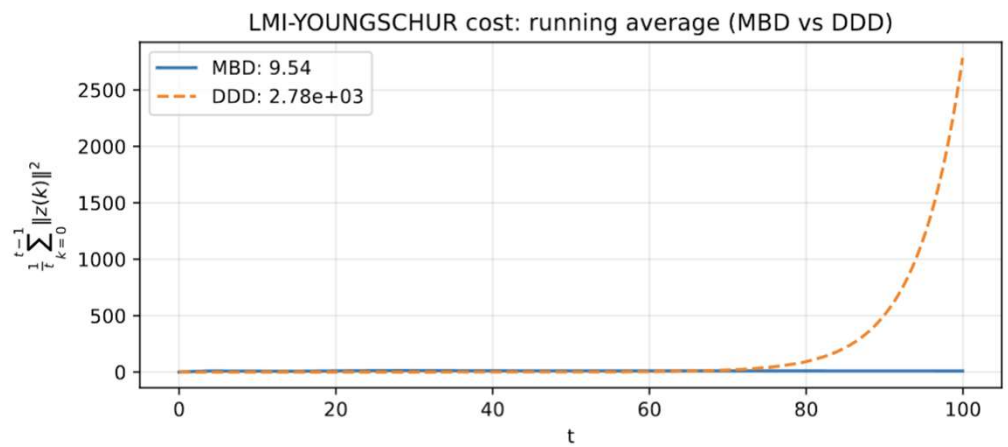
Young-Shur method

It fails to find a suitable solution.



$$\mathcal{S}_4 \subset \mathcal{S}_3 \subset \mathcal{S}_2 \subset \mathcal{S}_1 \subset \mathcal{S}_0$$





Error analysis / Failure cases

05.4

Discussion

07

Interpretation of results

07.1

Limitations of the method

07.2

Open questions

07.3

Conclusion

08

Key contributions (summary bullets)

08.1

Research roadmap (timeline diagram)

08.2

Appendix

09

Extra code/pseudocode

A.1

Mathematical derivations

A.2

Additional figures / experiments

A.3

Bibliography

10

References p1

SOTA Work

[1] R. Busetto, A. Lucchini, S. Formentin, and S. M. Savaresi, “Data-Driven Optimal Tuning of BLDC Motors With Safety Constraints: A Set Membership Approach,” *IEEE/ASME Trans. Mechatron.*, vol. 28, no. 4, pp. 1975–1983, Aug. 2023, doi: 10.1109/TMECH.2023.3269515.

[2] A. Colombo, R. Busetto, V. Breschi, M. Forgione, D. Piga, and S. Formentin, “In-Context Learning for Zero-Shot Speed Estimation of BLDC motors,” Apr. 01, 2025, *arXiv*: arXiv:2504.00673. doi: 10.48550/arXiv.2504.00673.

[3] R. Busetto, V. Breschi, M. Forgione, D. Piga, and S. Formentin, “In-context learning of state estimators,” *IFAC-PapersOnLine*, vol. 58, no. 15, pp. 145–150, 2024, doi: 10.1016/j.ifacol.2024.08.519.

[4] R. Busetto, V. Breschi, M. Forgione, D. Piga, and S. Formentin, “One controller to rule them all,” Nov. 10, 2024, *arXiv*: arXiv:2411.06482. doi: 10.48550/arXiv.2411.06482.

References p2

Correlated Work

[1] J.-C. Gamazo-Real, V. Martínez-Martínez, and J. Gomez-Gil, "ANN-based position and speed sensorless estimation for BLDC motors," *Measurement*, vol. 188, p. 110602, Jan. 2022, doi: 10.1016/j.measurement.2021.110602.

[2] G. Pillonetto, A. Aravkin, D. Gedon, L. Ljung, A. H. Ribeiro, and T. B. Schön, "Deep networks for system identification: A survey," *Automatica*, vol. 171, p. 111907, Jan. 2025, doi: 10.1016/j.automatica.2024.111907.

[3] M. C. Campi and S. M. Savaresi, "Direct Nonlinear Control Design: The Virtual Reference Feedback Tuning (VRFT) Approach," *IEEE Trans. Automat. Contr.*, vol. 51, no. 1, pp. 14–27, Jan. 2006, doi: 10.1109/TAC.2005.861689.

[4] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World," Mar. 20, 2017, *arXiv*: arXiv:1703.06907. doi: 10.48550/arXiv.1703.06907.

[5] L. Kirsch, J. Harrison, J. Sohl-Dickstein, and L. Metz, "General-Purpose In-Context Learning by Meta-Learning Transformers," Jan. 09, 2024, *arXiv*: arXiv:2212.04458. doi: 10.48550/arXiv.2212.04458.

[6] J. C. Gamazo-Real, E. Vázquez-Sánchez, and J. Gómez-Gil, "Position and speed control of brushless DC motors using sensorless techniques and application trends," *Sensors (Basel)*, vol. 10, no. 7, pp. 6901–6947, 2010, doi: 10.3390/s100706901.

[7] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-Real Transfer of Robotic Control with Dynamics Randomization," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, Brisbane, QLD: IEEE, May 2018, pp. 3803–3810. doi: 10.1109/ICRA.2018.8460528.

[8] L. S. Jr, F. Ruiz, and L. Fagiano, "SMGO-$Δ$: Balancing Caution and Reward in Global Optimization with Black-Box Constraints," Feb. 23, 2022, *arXiv*: arXiv:2202.11445. doi: 10.48550/arXiv.2202.11445.

[9] L. S. Jr, F. Ruiz, and L. Fagiano, "SMGO: A Set Membership Approach to Data-Driven Global Optimization," *Automatica*, vol. 133, p. 109890, Nov. 2021, doi: 10.1016/j.automatica.2021.109890.

[10] M. C. Campi, A. Lecchini, and S. M. Savaresi, "Virtual reference feedback tuning: a direct method for the design of feedback controllers," *Automatica*, vol. 38, no. 8, pp. 1337–1346, Aug. 2002, doi: 10.1016/S0005-1098(02)00032-8.

POLITECNICO MILANO 1863 | DEPARTMENT OF ELECTRONICS INFORMATION AND BIOENGINEERING

Thanks for the attention!

Contatti

Via Bassino – Edificio 20 – Secondo – Vano 031a
+39 3423066307
giuseppe.fiengo@polimi.it
Pagina Personale